MA3920 PROFESSIONAL INTERNSHIP

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Final Internship Report

Professional Internship / AY24-25

Acknowledgements

I would like to extend my deepest gratitude to Mr. Ng Christopher, my internship supervisor, for his unwavering support and guidance throughout the entire 20-week Professional Internship. His trust allowed me the autonomy to create the OHT Sensor System project from its conceptual phase to a final, manufacturable design, which significantly enhanced my professional and technical development.

I am sincerely grateful to Mr. Ian Carlos Salazar (Lead) and Mr. Mark Joseph Atienza, Senior Associate Engineers in the AMHS department. They provided continuous, hands-on mentorship, especially during the critical design phases involving sensor testing and the pivot to a modular, cost-efficient solution. Their engineering feedback was instrumental in delivering a practical project deliverable.

Special thanks to Mr. Tay See Hong for his assistance in coordinating vendor communications and procurement logistics for the sensor mount fabrication. I also thank Mr. Tan Tiah Long for his earlier detailed tours of Fab 7, 7G and 7H, which provided the foundational context necessary for designing a cross-fab solution.

The entire AMHS team including Mr. Nay Tun Naing and Mr. Alan Ho, deserves recognition for their continuous support in navigating technical documentation and internal theories.

Finally, I thank all GlobalFoundries staff who contributed to my learning, including the Facilities team for the job shadow opportunity, all of which enriched my understanding of the broader semiconductor ecosystem.

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1. Introduction

1.1 Company Overview

GlobalFoundries (GF) is a leading global semiconductor foundry that provides advanced manufacturing services for integrated circuits. Operating on a pure-play foundry model, GF manufactures chips designed by other companies. The company fosters a culture of innovation, collaboration and continuous improvement, with a strong emphasis on safety, quality and operational excellence.

GF offers a wide range of semiconductor manufacturing technologies, including radio frequency (RF), analog and embedded memory solutions. Its services span from design enablement to wafer fabrication and testing. GF serves global markets, with key customers in automotive, mobile, IoT and data centre sectors. Suppliers include equipment manufacturers and raw materials providers while competitors include TSMC, Samsung Foundry, UMC and Intel.

GF’s competitive edge lies in its focus on specialized process technologies and advanced packaging. At the heart of its Asia-Pacific operations, GF Singapore (APAC headquarters) operates Fab GIGA+ and Fab 7, 7G and 7H, which manufacture chips on both 200mm and 300mm wafers, supporting technologies from 350nm to 40nm.

GF APAC also fosters a supportive and inclusive work environment that emphasizes both professional development and employee well-being. The company offers a variety of employee engagement initiatives, including additional training, weekly sports activities, cultural celebrations, and team bonding events. GF Singapore has also been recognized as the Employee Experience Champion of the Year at the 2025 Employee Experience Awards in Singapore, winning multiple awards for its excellence in leadership, learning, engagement, and talent acquisition.

In addition, GF Singapore supports talent development through paid internships, traineeships, and work-study programs in collaboration with local institutions. These initiatives provide hands-on experience in semiconductor manufacturing and open pathways to full-time employment, reinforcing GF’s commitment to nurturing future talent in the region.

2. Internship Background

2.1 Role of an Engineer in the AMHS Department.

The Automated Material Handling System (AMHS) is a vital infrastructure within the semiconductor fab that ensures the smooth and efficient movement of materials. It is responsible for transferring FOUPs (Front Opening Unified Pods), which carry wafers, between different process tools in a highly controlled cleanroom environment. AMHS helps reduce manual handling, minimises contamination risk and improves production flow.

One of the key components of the AMHS is the Overhead Hoist Transport (OHT) system. These are rail-guided vehicles that run on tracks suspended from the ceiling. OHTs pick up and deliver FOUPs based on instructions from the Material Control System (MCS). Because wafers are extremely sensitive to particles, vibrations and delays. OHTs must also operate with high accuracy and reliability. Any fault or downtime in the OHT system can disrupt the production schedule and reduce overall fab efficiency.

To support daily operations and ensure accountability, I attended the AMHS department's 8:00 AM daily pass-down meetings. These meetings reviewed the previous day's Preventive Maintenance (PM) and Corrective Maintenance (CM) activities on the OHT systems. Line support technicians reported the specific issues encountered during their shifts, allowing for accountability when a repaired OHT returns with further issues. This detailed tracking ensures that the engineer who performed the previous maintenance can be identified and consulted, promoting responsibility and transparency in maintenance activities. These updates are used by senior engineers to document maintenance history in the AMHS data system and to identify any components that require urgent replacement or restocking.

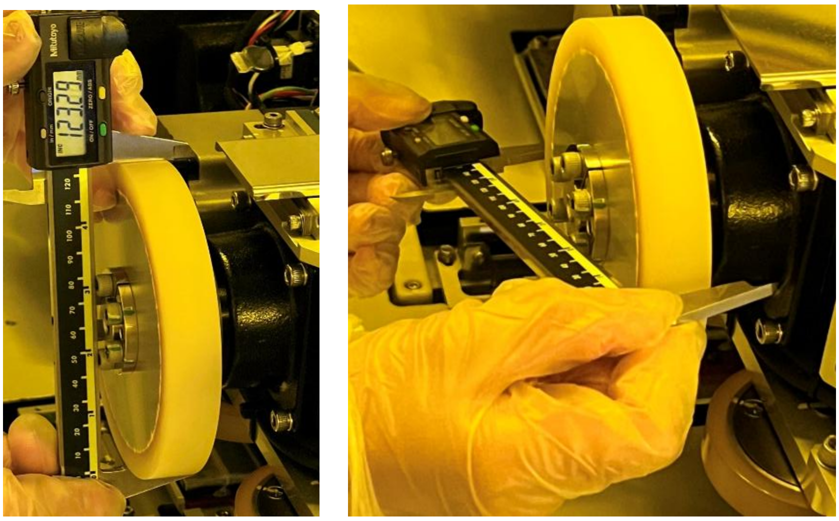
The meeting also provided visibility on the total OHT utilization for Fab 7, 7G and 7H enabling trend analysis. Anomalies in utilization rates can affect overall production, as reduced OHT activity may indicate delays or inefficiencies in wafer transport. Additionally, AMHS oversees not only OHTs but also stockers, which function as storage systems for both empty and loaded FOUPs awaiting pickup. The performance of both systems contributes to the smooth operation of the fab.

The maintenance and performance of OHTs are critical to achieving high FOUPs move and meeting fab productivity goals. That is why improvements to OHT monitoring such as predictive maintenance systems, are important for reducing errors and unplanned downtimes in wafer transport operations.

3. Main Project: Sensor System for OHT Wheel Monitoring

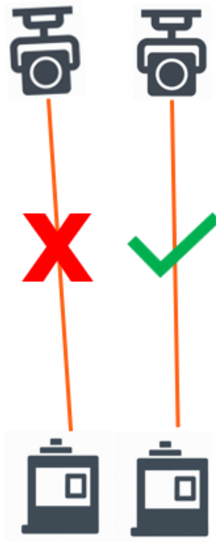
3.1 Problem Description and Project Rationale

The project addresses the unreliability of the current manual maintenance process used to check OHT wheel wear. Technicians currently use a vernier caliper to measure wheel diameter, a method highly prone to human error. Small inconsistencies in pressure or measurement angle can lead to inaccurate and inconsistent data, evidenced by impossible readings (like recorded diameter appearing to increase over time), rendering the collected information unreliable.



Using Vernier Caliper for Measurement

A critical issue is that the manual measurements cannot reliably detect uneven wear especially when one side of the OHT's wheels wears down more than the other. This uneven wear is a serious problem as it will cause the OHT to lean one side slightly. This minor angle deviation can disrupt the FOUP look-down sensor's alignment. This misalignment leads to failed pick-ups or transport delays, forcing the OHT to repeatedly try and readjust its position. Such issues disrupt the wafer flow, reduce fab efficiency and highlight the urgent need for a more reliable measuring system.



OHT Misalignment

3.2 Project Objectives and My Role

To address these shortcomings, the AMHS department aims to implement an automated system using sensors to obtain highly accurate and consistent measurements. The core objective is to leverage this reliable data to integrate machine learning (ML) techniques for predicting the Remaining Useful Life (RUL) of the OHT wheels. This predictive maintenance approach will allow us to schedule maintenance proactively, therefore improving data integrity and minimizing unplanned downtimes.

My role in this project involves:

- Analyzing the current manual maintenance process.
- Researching and identifying suitable sensor technologies.
- Developing a prototype system, which involved designing the sensor mount.
- Exploring predictive analytical models.

3.3 Project Outline and Current Status

The automated measurement system is being developed by a project team led by two senior associate engineers, Ian and Mark, and supported by myself. Our long-term goal is to replace the current manual measurements with an accurate, consistent sensor system and enable full predictive maintenance capabilities.

The project is currently in its first phase: ideation, sensor procurement and prototype design. I began by conducting on-site observations of the OHT system and maintenance workflow. These observations were crucial for defining technical requirements, including necessary measurement precision, optimal sensor placement and overall implementation feasibility within operational and environmental constraints.

We then held planning discussions and conducted research on suitable sensor technologies from vendors such as Keyence and Omron, evaluating options including ultrasonic, Light Detection and Ranging (LiDAR) and vision-based systems. After thoroughly evaluating factors like precision, compatibility, size, ease of integration and cost, LiDAR was deemed the most cost-effective solution that still met our accuracy and requirements.

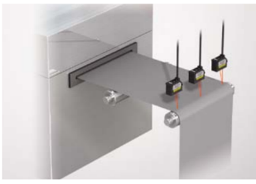
We developed proposal slides for our supervisor, Mr. Ng Christopher, to justify this selection, detailing the system's placement, expected precision and implementation feasibility within the project timeline. These proposal materials are regularly updated to ensure project alignment and are intended to convince decision-makers of the project's value. Future iterations of the proposal will include a Return on Investment (ROI) analysis to highlight potential cost savings and long-term operational efficiency gains.

Once the sensor is tested and validated, the next steps will involve collecting real-time data and comparing it with historical records to confirm accuracy. This validated dataset will serve as the foundation for training the machine learning model to predict wheel RUL. The final solution will include a dashboard or alert system to notify engineers when an OHT approaches its maintenance threshold.

Intelligent Laser (IL) Sensor

IL-300

Sheet material thickness detection



This device constantly monitors thickness differentiation of sheet material. The multi-point head allows you to detect irregularities in the thickness of edges and in the bodies of materials.

Control sheet roll diameter

Control feed speed and tension rolling with constant monitoring of sheet diameter during rolling and unrolling processes.

Model	IL- 300
Reference Distance	300 mm
Measurement Range	160 to 450 mm
Sampling Rate	0.33/1/2/5 ms (4 levels available)
Operational Status Indicator	Laser emission warning indicator: Green LED, Analogue range indicator: Orange LED, Reference distance indicator: Red/Green LED
Material	Housing material: PBT, Metal parts: SUS304, Packing: NBR, Lens cover: Glass, Cable: PVC
Weight	Approx. 135g

Diffuse-reflective	Line beam	300 ± 150 mm; 150 mm 450 mm	4 µm	0.2 m	ZP-LS300L 0.2M
				2 m	ZP-LS300L 2M
	Spot beam	300 ± 150 mm; 150 mm 450 mm	4 µm	0.2 m	ZP-LS300S 0.2M
				2 m	ZP-LS300S 2M
	Line beam	600 ± 400 mm; 200 mm 1000 mm	14 µm	0.2 m	ZP-LS600L 0.2M
				2 m	ZP-LS600L 2M
	Spot beam	600 ± 400 mm; 200 mm 1000 mm	14 µm	0.2 m	ZP-LS600S 0.2M
				2 m	ZP-LS600S 2M

<u>Sensor Choices</u>

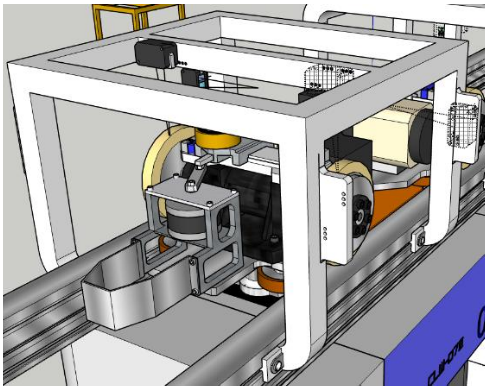
3.4 Design Process and Evolution

The main task for me was to design the physical mount to hold the sensor. This design changed many times based on new information and feedback.

1. **Initial Research:** The team started by observing the OHT movement to select the location to place the system and to account for the space constraint up on the track. We ended up selecting Omron LiDAR sensor and contacted the vendor to get a T-loan unit to begin testing.

2. **First Hurdle (Universal Design):** My first design was for Fab 7 only. However, the team realized we needed a “universal” design that could also work in Fab 7G and 7H to save manufacturing costs.

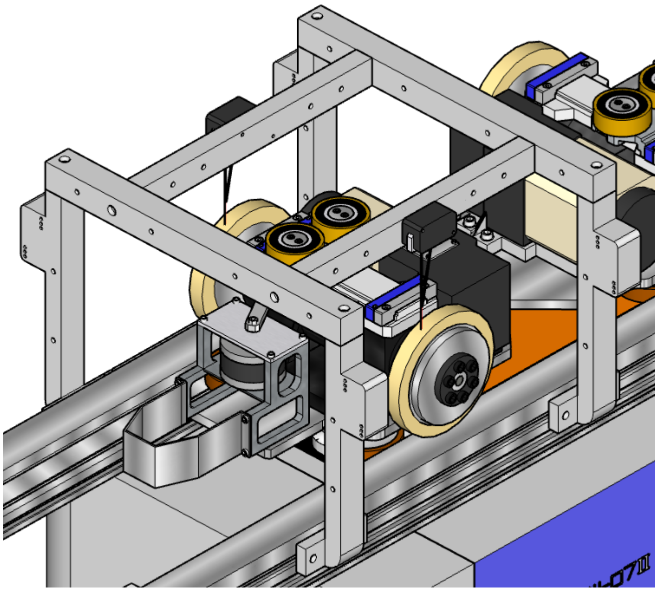
3. **Critical Discovery (Sensor Placement):** During on-the-ground testing, we made a key discovery. The initial concept which was designed based on the perpendicular placement of the sensor to the wheel, gave fluctuating and inaccurate results. We did further test and found out that the sensor must be placed parallel to the OHT wheel to get accurate and consistent height measurements. It could be due to how the light bounces off when the OHT approaches the sensor which gave the inconsistency.



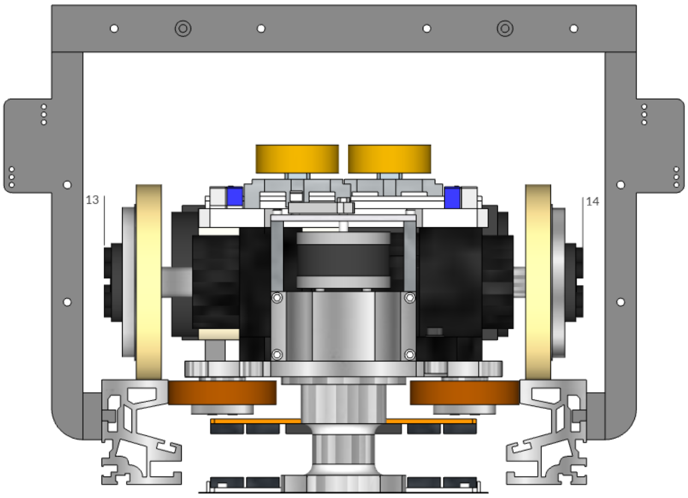
<u>Sensor (Black) is Perpendicular to the Wheel</u>

4. **Major Redesign (Feedback):** This discovery forced a complete redesign. I created a new “Roman numeral II” one-piece design to stably hold the sensor in the precise parallel position.

5. **Verification:** Before getting approval, I had to prove this new design was aligned and within the space constraints. I used SketchUp to build a 3D model of the entire OHT and its track. To create the model accurately, I would have to access the 2D draft of each drawing. I then put my virtual frame model into the simulation. This let me check all the tolerances and make sure there were safe gaps. This was critical to prevent the frame from ever hitting the OHT, which would be a serious fab incident. This verification allows me to adjust the frame according to spatial constraints.



Roman II Design

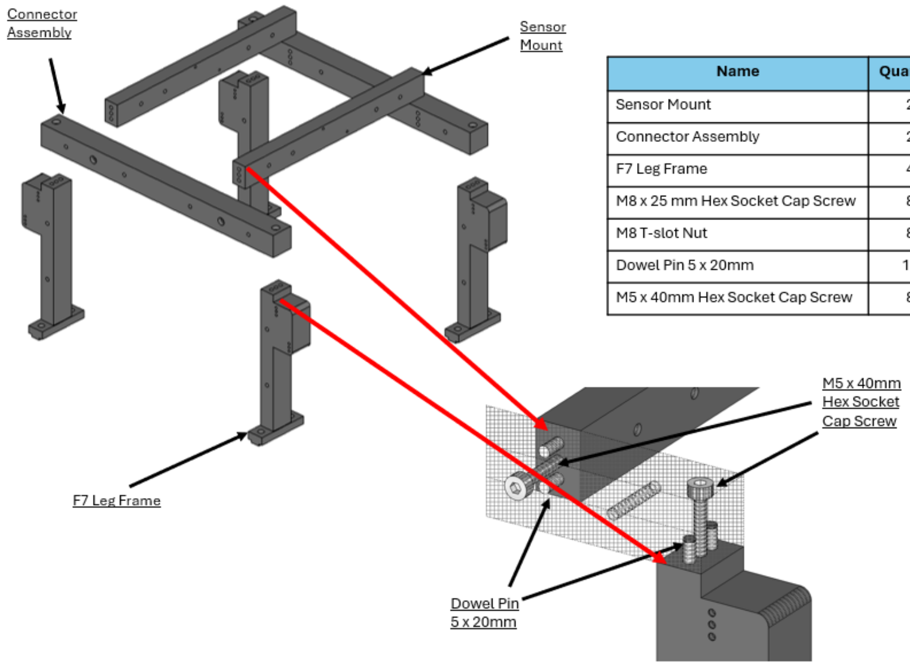


Frame-Wheel Safety Tolerance

6. **Final Modular Design:** The “II” design was approved, but my supervisor gave new feedback: make it into pieces to save manufacturing costs. He shared with me that manufacturers would have to use a large piece of aluminum to carve out the one-piece frame design which will cost a lot. Split pieces design would mean that smaller aluminum block would be used. This led me to develop a modular approach. I managed to split the design into 8 -part design. However, I could further simplify the design into a final, highly cost effective 4-part modular assembly.

This final design was the key breakthrough. It consisted of four unique components:

- A Sensor Mount (holds the sensor)
- A Connector Assembly (bridges the support and sensor mount)
- A Fab 7 Support Frame (the legs for the connector assembly)
- A Fab 7G/H Support Frame (the legs for the connector assembly)



Name	Quantity	MATERIAL
Sensor Mount	2	ALUMINIUM
Connector Assembly	2	ALUMINIUM
F7 Leg Frame	4	ALUMINIUM
M8 x 25 mm Hex Socket Cap Screw	8	Stainless Steel
M8 T-slot Nut	8	Stainless Steel
Dowel Pin 5 x 20mm	16	Stainless Steel
M5 x 40mm Hex Socket Cap Screw	8	Stainless Steel

F7 Sensor Mount Assembly

3.5 Key Challenges and Solutions

I faced several key challenges during the design process.

- **Challenge 1: Design for Manufacturing (DFM)**
  - My supervisor taught me that a design is not just a drawing. The one-piece "II" design was rejected because of the possible excessive manufacturing cost. From his experience, he knows that vendors will carve it from a single aluminum block, and it will lead to a waste of material and require a more complex machining process, pushing it far outside our budget.
  - <u>Solution:</u> This forced me to learn and apply DFM. The 4-part modular design was a direct result of this. It was much cheaper to produce four smaller, simpler parts.
- **Challenge 2: Precision Alignment**
  - This new 4-part design created a new engineering challenge. Ensuring high-precision alignment when assembling the separate parts. Slight misalignment when assembling could lead to sensor mis-detecting the OHT wheel.
  - <u>Solution:</u> I designed a meticulous connection system. Dowel pins (e.g. 5mm x 10mm stainless steel pins) were used to locate and lock the pieces in the exact right position. Then, hex socket cap screws (e.g., M5 x 40mm and M8 x 20mm) were used to fasten the components securely for stability.
- **Challenge 3: Lack of CAD Software**
  - My biggest technical challenge was that the company did not have a dedicated CAD application like SolidWorks. I had to use the free version of SketchUp, which was difficult due to the workaround that I had to do to get a precise model out. It also does not have a 2D drafting functionality.
  - <u>Solution:</u> I taught myself to use SketchUp for 3D modeling through YouTube videos on certain way of creating a model. For the 2D technical drawings, I developed a workaround. I took screenshots from SketchUp and imported them into Microsoft PowerPoint and Paint. In PowerPoint, I could align and size the views properly and add the correct dimensions to create the final plans for the manufacturer.

3.6 Project Status: Phased Procurement and Risk Mitigation

As of the end of the internship, I had successfully completed the final 4-part modular design and its 2D drawings. I created a full proposal with layouts, a logistics list and assembly plans. My supervisor approved the design but suggested a strategic decision to mitigate risk.

- **The Action:** Instead of ordering the full quantity, we began a phased procurement. We requested for sample batch of the 4 unique aluminium pieces for us to test the support frame on each of the fab.

- **The learning point:** This was a vital lesson. This strategy allows the team to physically install and test the real frame on the OHT track across different fab as Fab 7 track is different compared to Fab 7G and 7H's. We can check the fit and the sensor's data stability. If any slight adjustments are needed, I can adjust the 2D technical drawing before our department fully commits to the full, expensive order, preventing costly mistakes.

4. Other Activities and Learning Experiences

Besides my main project, I was involved in other activities that helped my learning.

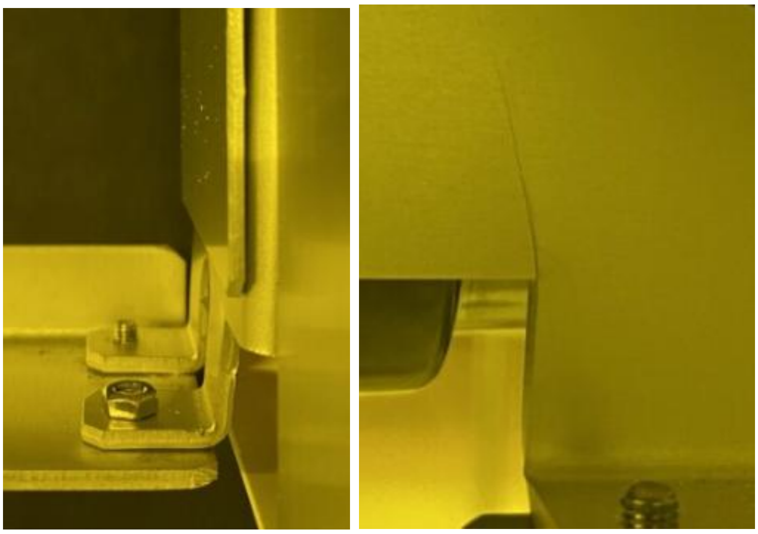
4.1 Technical and Operational Exposure

- I assisted line support technicians during OHT maintenance, which gave me a practical look at how wear and tear affects the system.
- I attended a 2-day job shadow with Fab 7's Facilities Department. I was amazed at how vast and big the Fab's ecosystem is. I toured the Fab, Sub Fab and Roof and learned about the complex systems that support the cleanroom such as air conditioning, particles stabilization, chemical mixing and fire extinguishing systems.

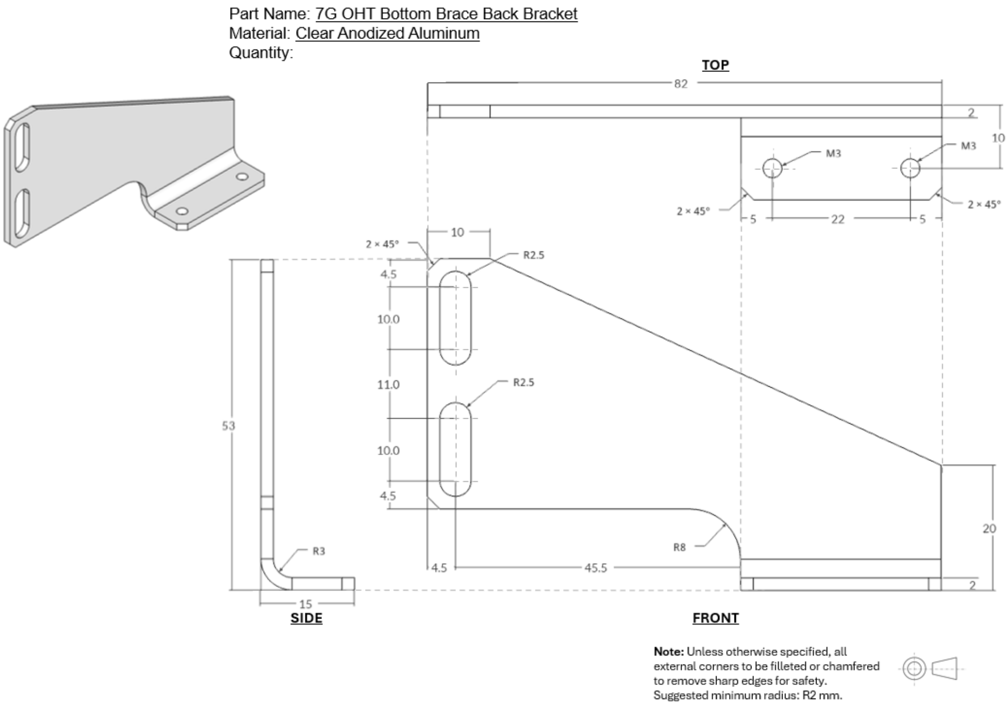
4.2 Other Tasking: OHT Brace Bar Bracket Redesign

I helped solve a problem with the OHT Brace Bar brackets. The original brackets kept breaking because they were made by bending a flat metal plate into a sharp 90° bend. The constant shaking and movement of the OHT created too much stress at this sharp corner, causing the bracket to fail.

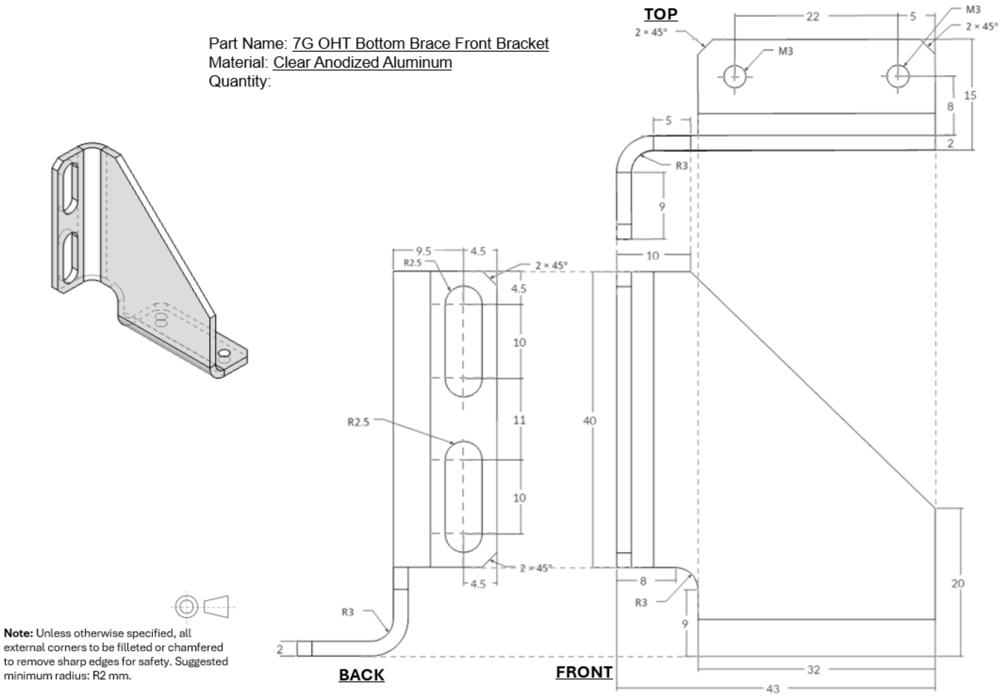
My job was to design a solid replacement bracket. The key design fix was to make sure the new bracket was machined from a solid block of aluminum instead of being cut out into pieces and bend at specific parts. I also added a 3 mm radius curve (fillet) on the critical edges. This curve helps spread stress, making the new structure much stronger and stopping it from breaking under vibration. This task taught me how small design choices, like adding a curve, can fix big reliability problems.



Bracket that has Fault



Fab 7G OHT Bottom Brace Back Bracket



Fab 7G OHT Bottom Brace Front Bracket

5. Reflections & Conclusion

5.1 Key Learning Points

This 20-week internship was an eye-opening experience.

- I learned that real-world engineering is about trade-offs. A design must balance performance, cost, manufacturability and long-term maintenance.
- The technical pivot from a perpendicular to a parallel sensor placement was found out through real-world testing, this proved that physical validation is an important necessary step, and we shouldn’t just rely on theoretical assumptions.
- I gained technical skills in CAD design using SketchUp, but more importantly, I learned how to be resourceful. When I faced a tool limitation (no 2D drawing functionality), I found my own solution using PowerPoint to get the job done.
- I learned the importance of teamwork and communication. Sharing ideas with my team and getting constant feedback from my supervisor was key to improving the design.

5.2 Personal Shortcomings

I also faced areas where I could have done better.

- My biggest challenge was time management related to the project timeline. I underestimated how long certain critical phases would take.
- For example, the detailed design phase, where I worked to reduce the component count for cost-saving, took extra time. This effort, while necessary, pushed back the fabrication timeline.
- I also underestimated how long it takes to get internal approvals for budget and designs. I learned that I should have pushed myself to finalize these submissions sooner to account for these administrative delays. This experience taught me the importance of managing timelines more effectively, especially when faced with unexpected delays.

5.3 Final Conclusion

My Professional Internship at GlobalFoundries ended on a very positive note. In the final weeks, my main project achieved its most important milestone: the 4-part modular sensor mount design was fully approved by my supervisor for manufacturing.

After I submitted the final 2D drawings and logistics list, the design was officially sent out to the vendor for a price quotation and timeline estimation for the sample batch.

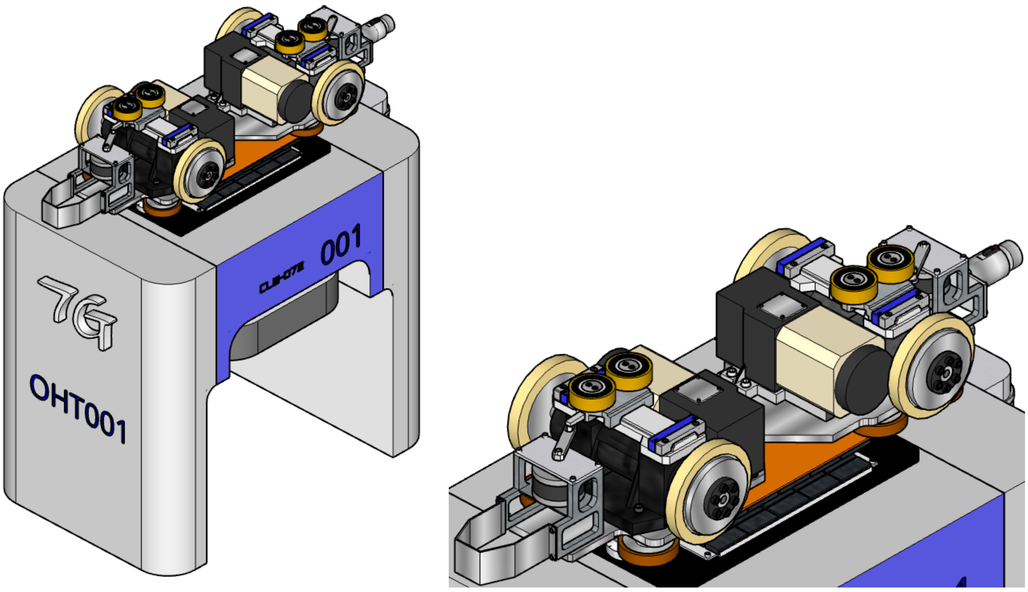
Looking back, this internship was a valuable, hands-on journey. I successfully solved a complex problem, the need for an automated wheel measurement system and delivered a complete, manufacturable solution. I learned practical skills that school does not teach, such as Design for Manufacturing (DFM) to reduce costs, risk mitigation by ordering a sample batch and the importance of virtual checks. I also learned to be resourceful, finding my own way to create technical drawings using PowerPoint when I did not have the right software.

This experience has taught me how to be a professional, accountable and practical engineer. I am proud to have created a final design that will help my project team to move forward with the predictive maintenance goals.

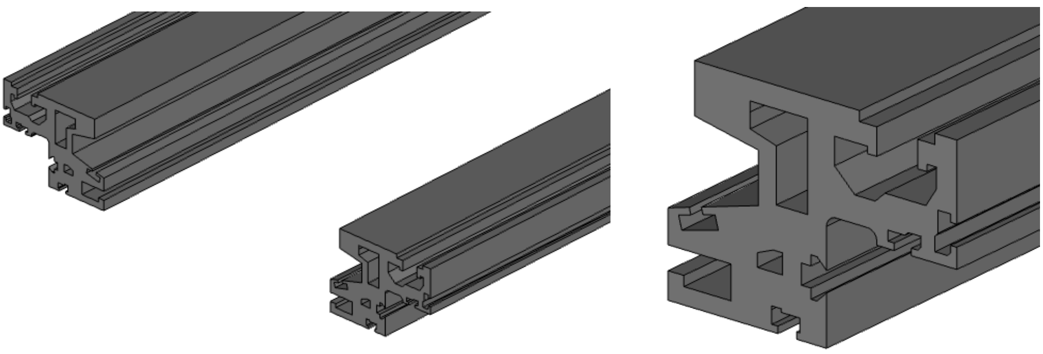
6. Appendices

Note on Visuals: To comply with GlobalFoundries' confidentiality guidelines, visual materials such as CAD screenshots, internal data tables, or site-specific images have either been excluded or edited to remove any proprietary or sensitive content. Visual assets in this report are conceptual in nature and intended only to support general understanding of the project scope. Internal visuals are stored securely within the company's internal systems and are available only to authorised personnel.

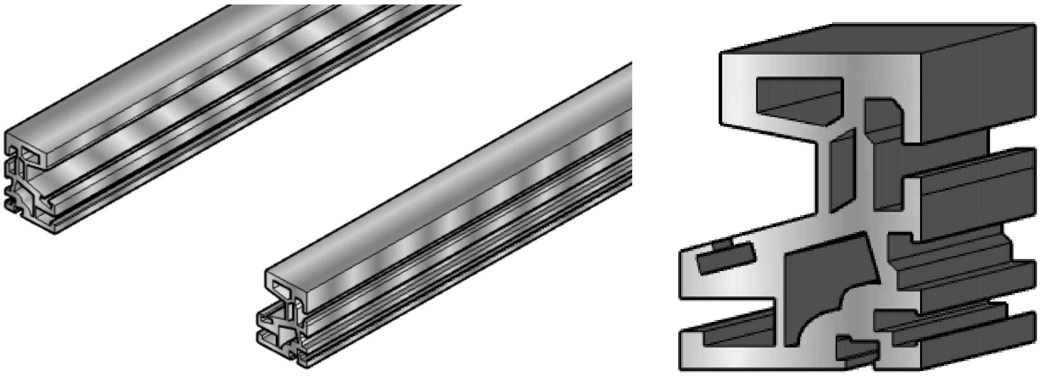
6.1 Appendix A – 3D Design Iterations



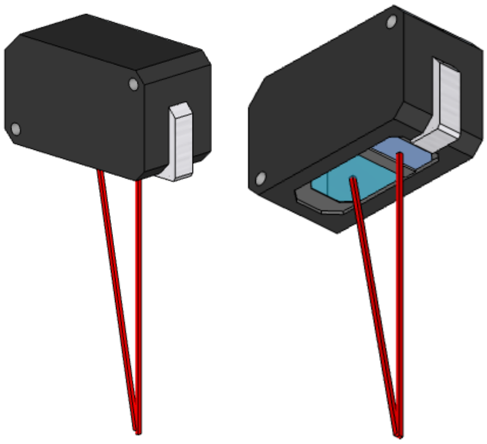
Overhead Hoist Transport



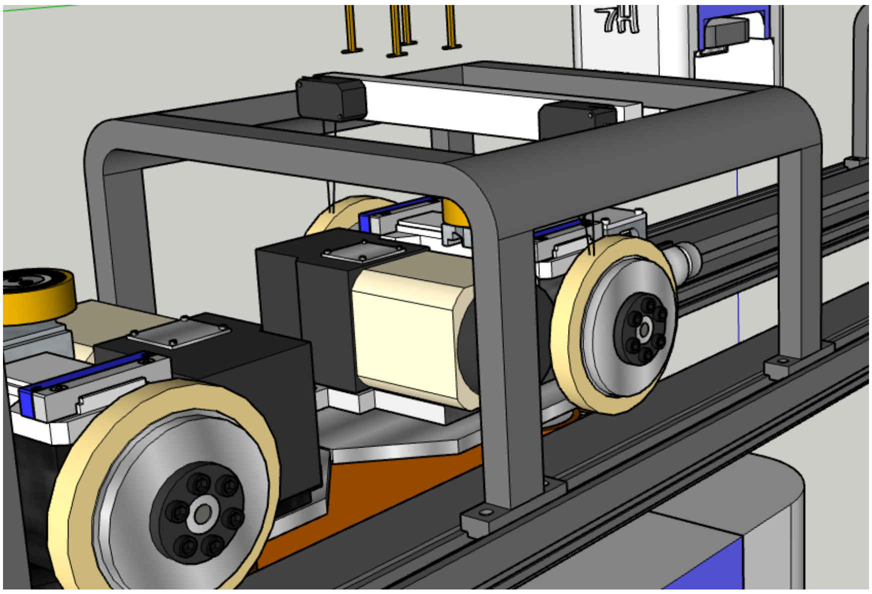
F7 OHT Track



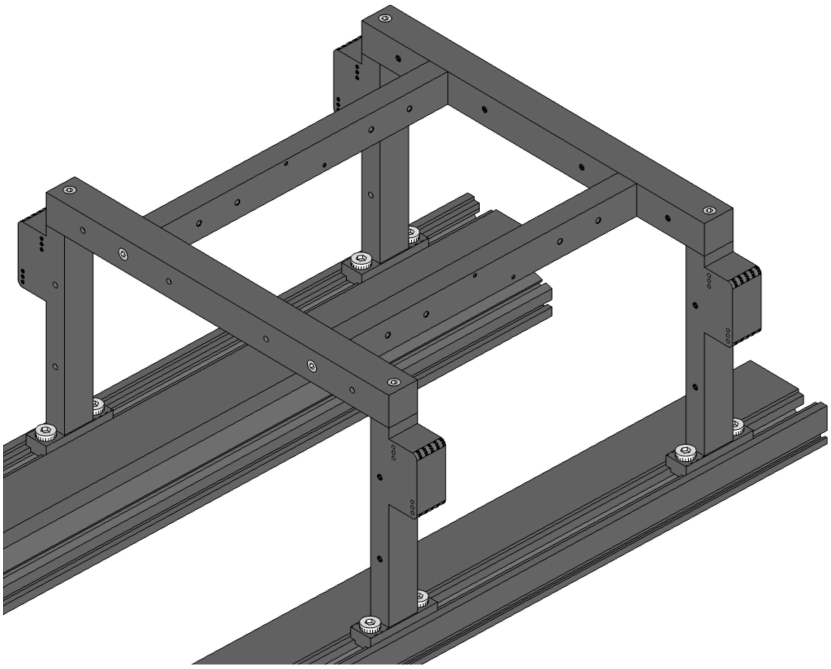
F7G/H OHT Track



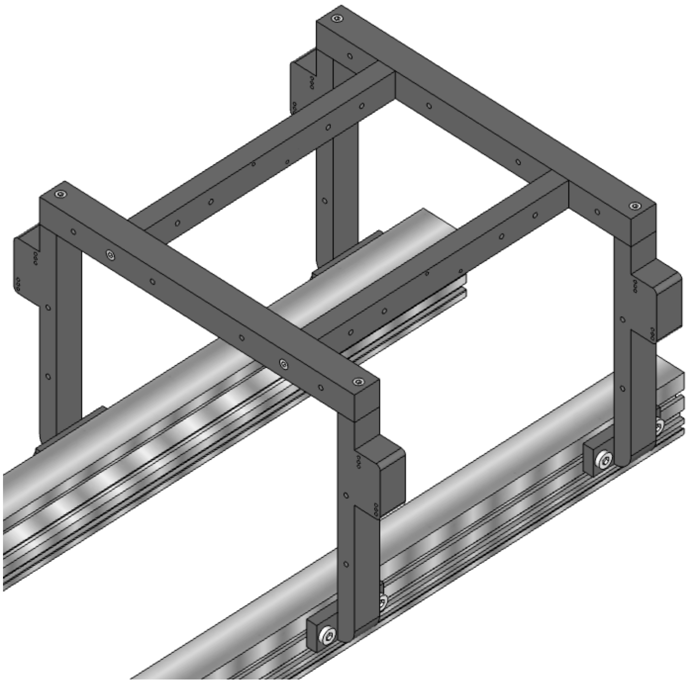
LiDAR Sensor



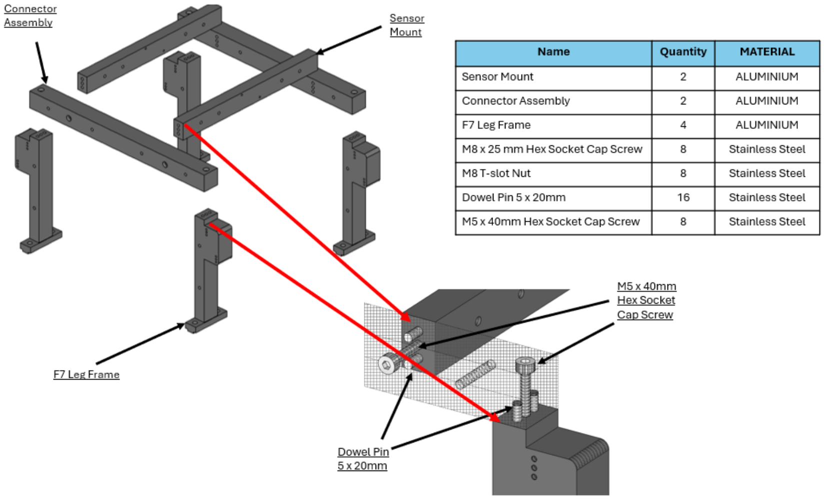
F7 Frame (1 Piece Design)



F7 Frame Assembly (4 Unique Piece Design)

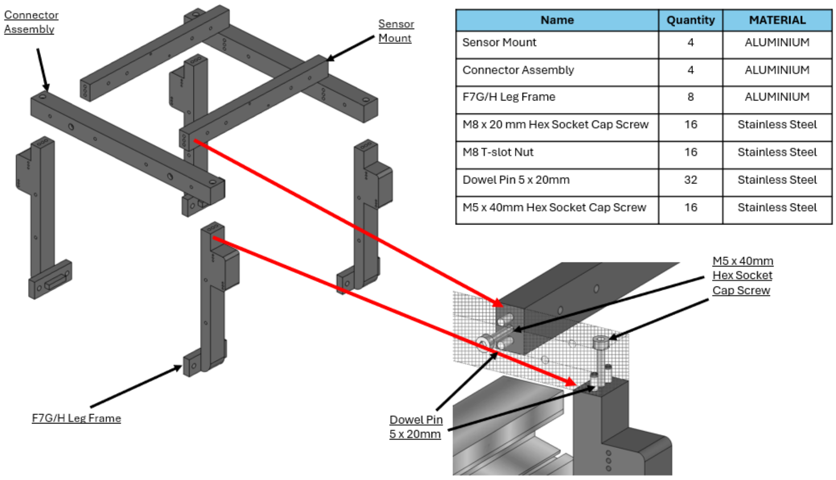


F7G/H Frame Assembly (4 Unique Piece Design)



Name	Quantity	MATERIAL
Sensor Mount	2	ALUMINIUM
Connector Assembly	2	ALUMINIUM
F7 Leg Frame	4	ALUMINIUM
M8 x 25 mm Hex Socket Cap Screw	8	Stainless Steel
M8 T-slot Nut	8	Stainless Steel
Dowel Pin 5 x 20mm	16	Stainless Steel
M5 x 40mm Hex Socket Cap Screw	8	Stainless Steel

F7 Frame Assembly Overview



Name	Quantity	MATERIAL
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Connector Assembly	4	ALUMINIUM
F7G/H Leg Frame	8	ALUMINIUM
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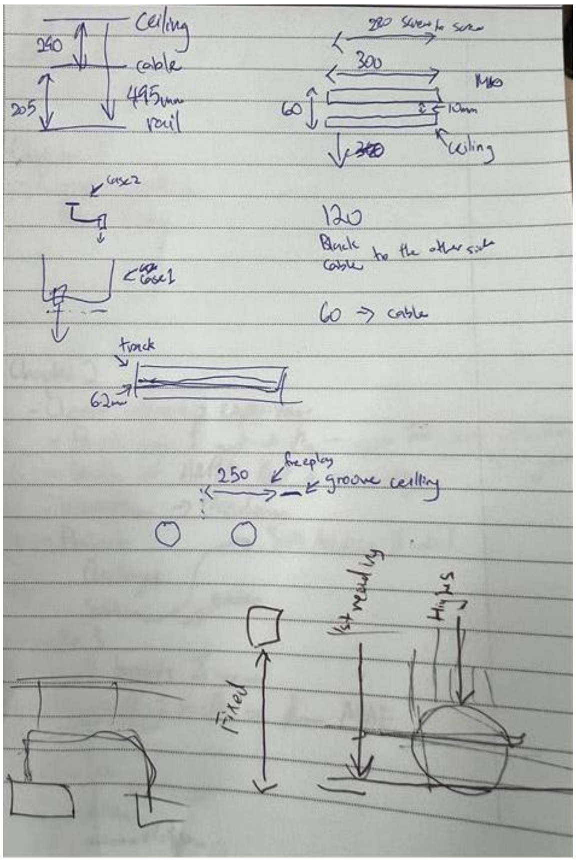
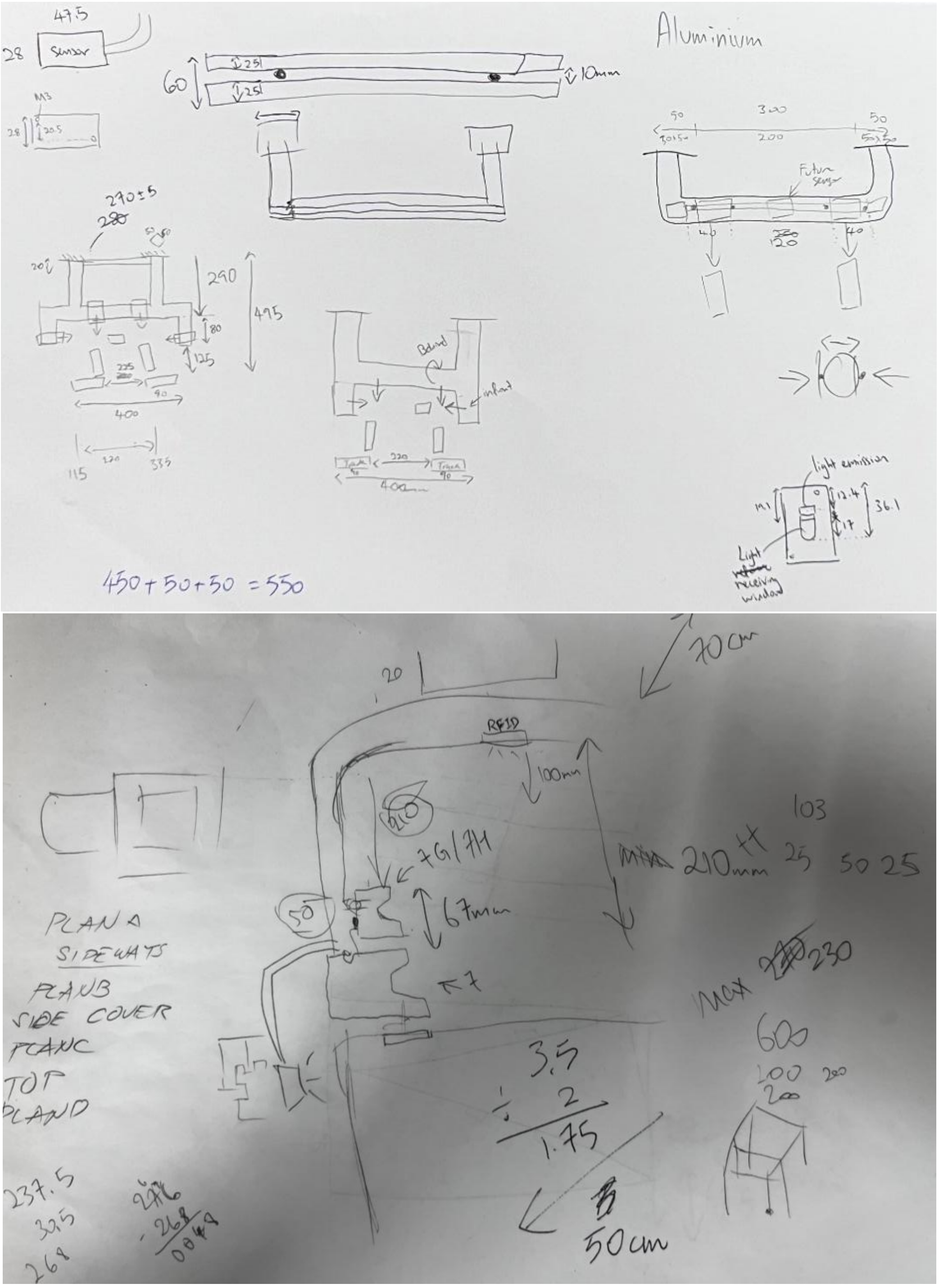
M8 T-slot Nut	16	Stainless Steel
Dowel Pin 5 x 20mm	32	Stainless Steel
M5 x 40mm Hex Socket Cap Screw	16	Stainless Steel

F7G/H Frame Assembly Overview

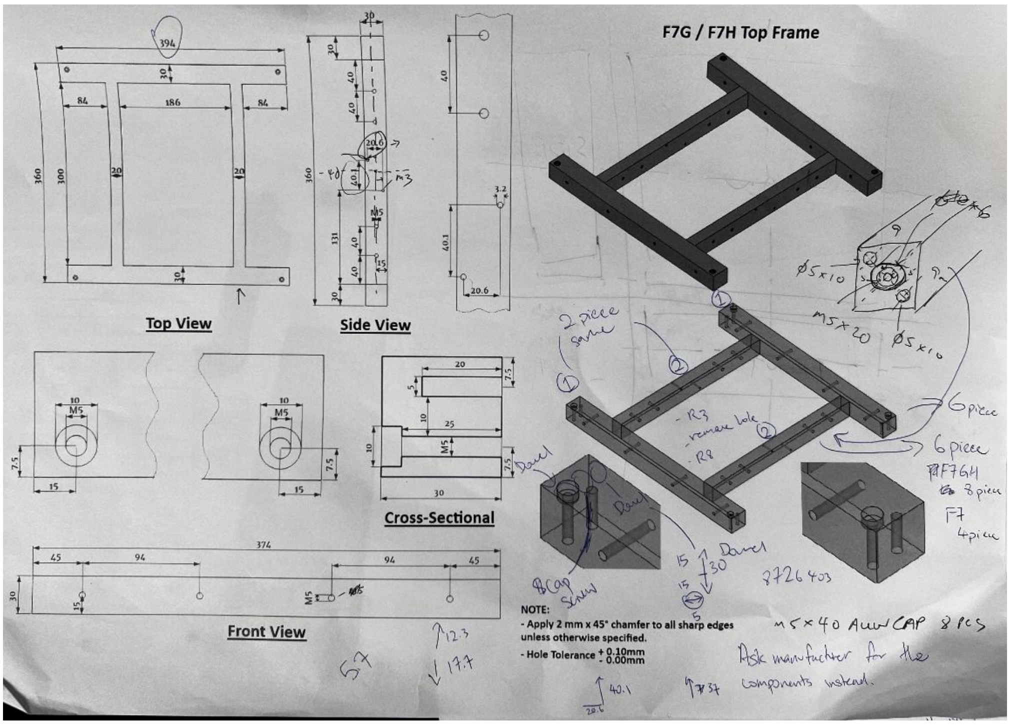
6.2 Appendix B – Engineering Notes: Measurements and Layout Discussion



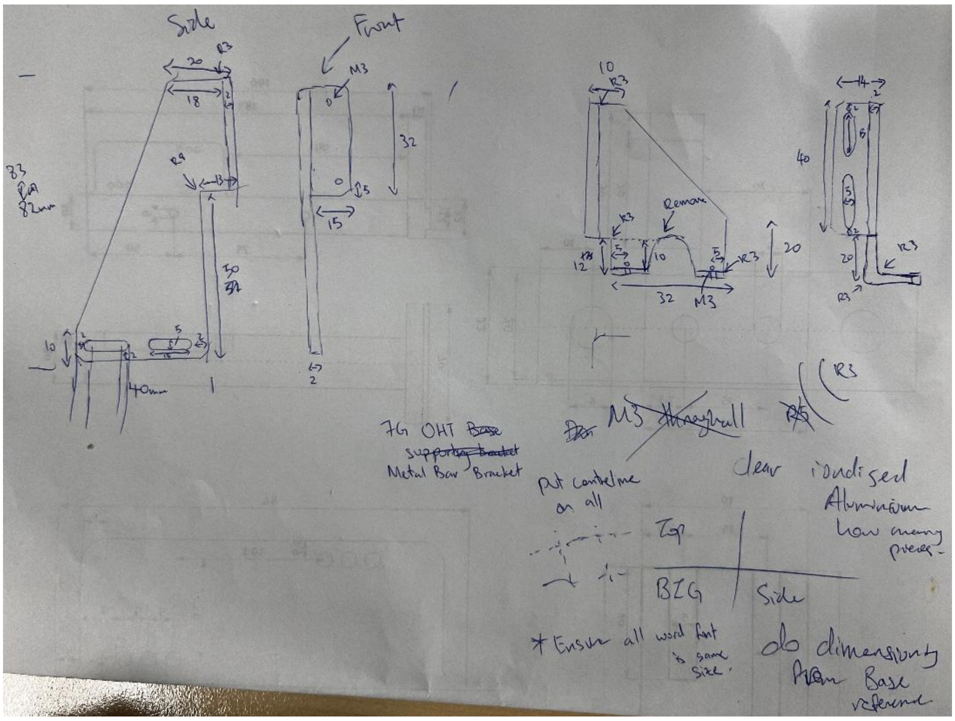
Measurement taken in the Cleanroom Maintenance Room



Sketches of Possible Frame Design

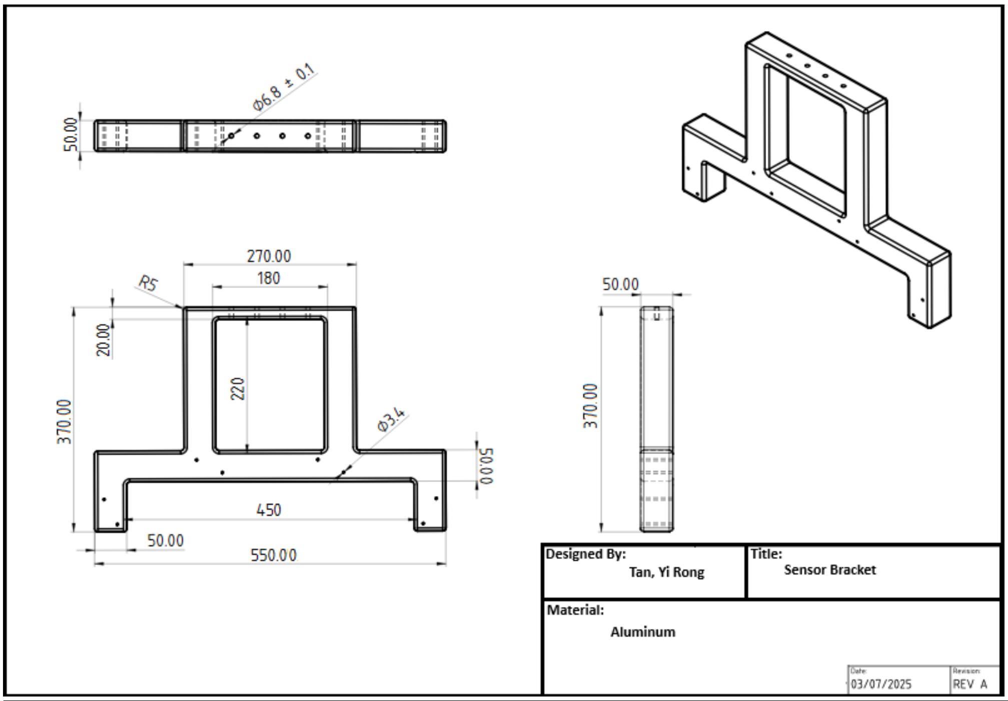


Feedback from Supervisor on What to Change/Improve



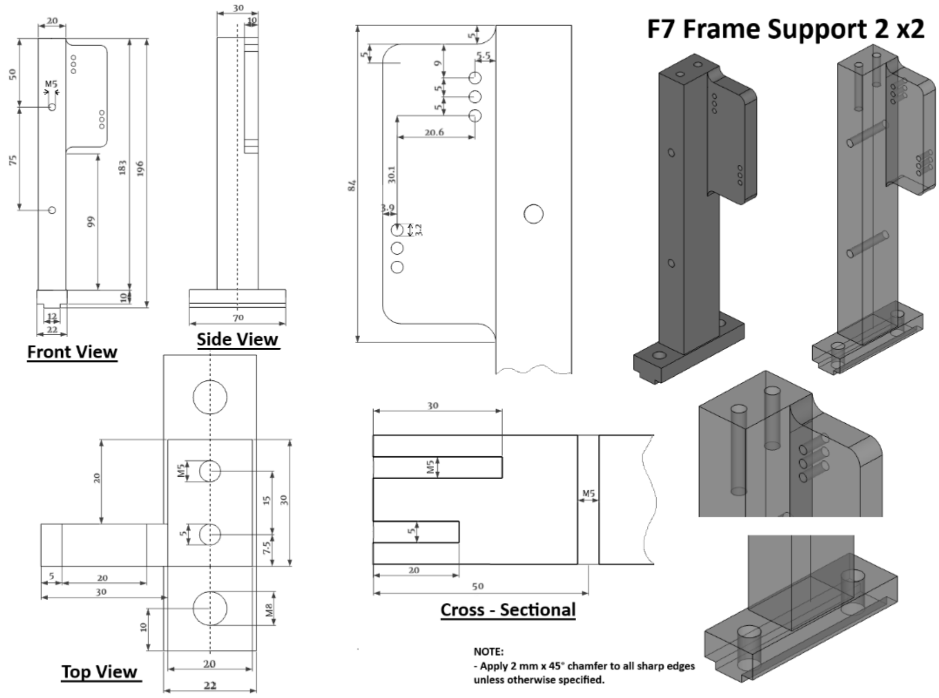
Measurements of Faulty Bracket Sketched Out

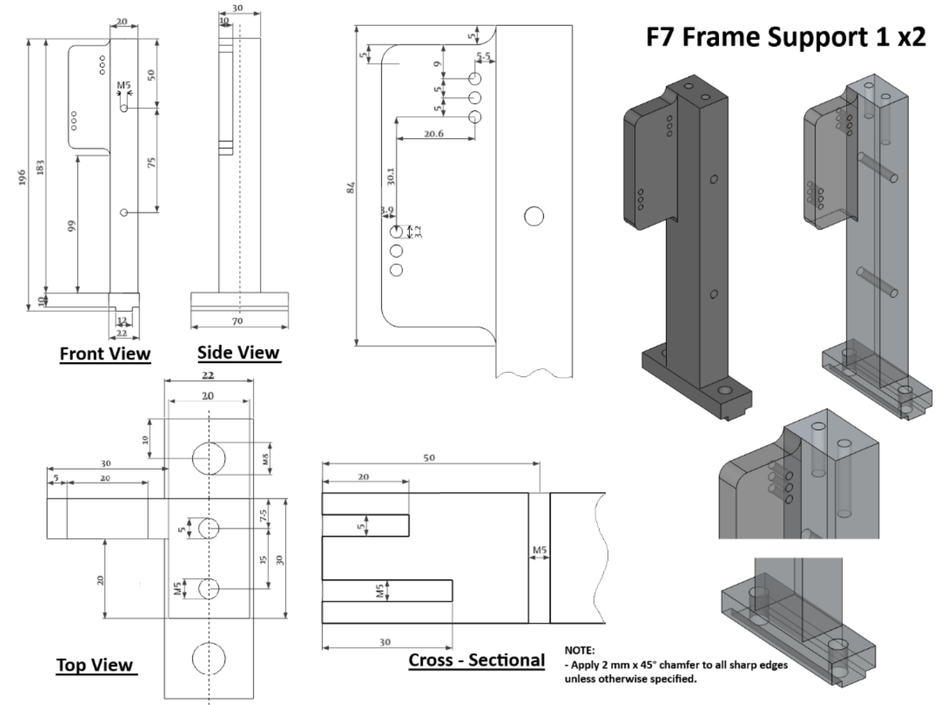
6.3 Appendix C – 2D Technical Drawings

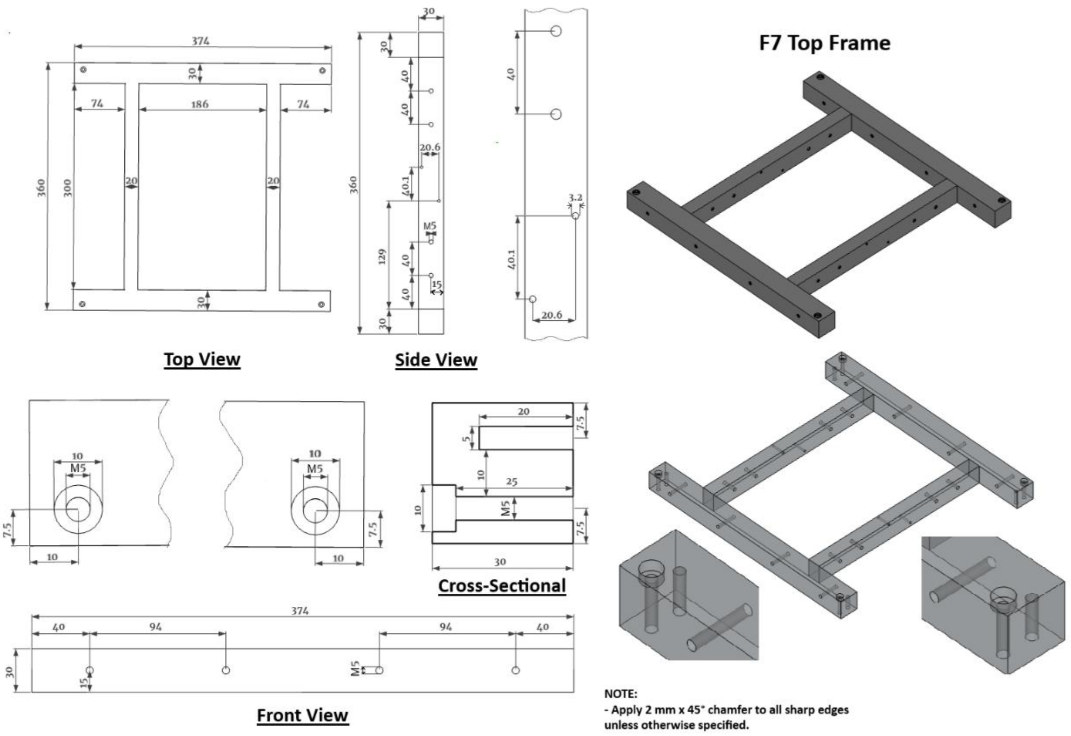


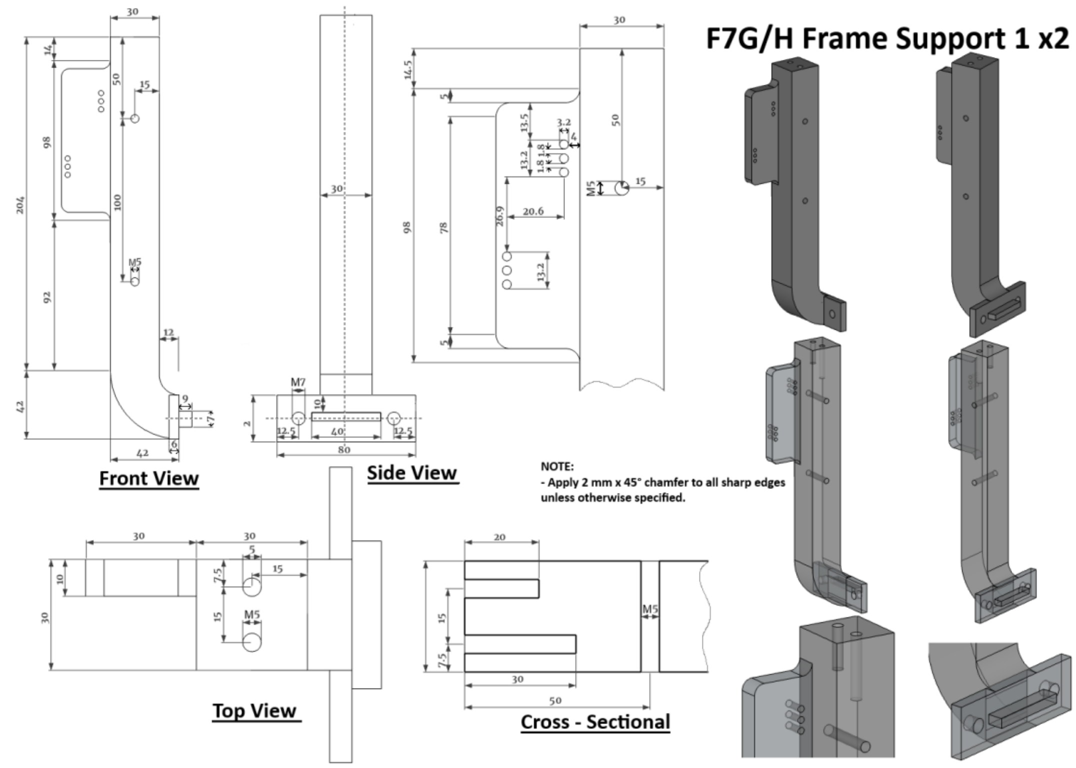
Initial Ceiling Attached Design

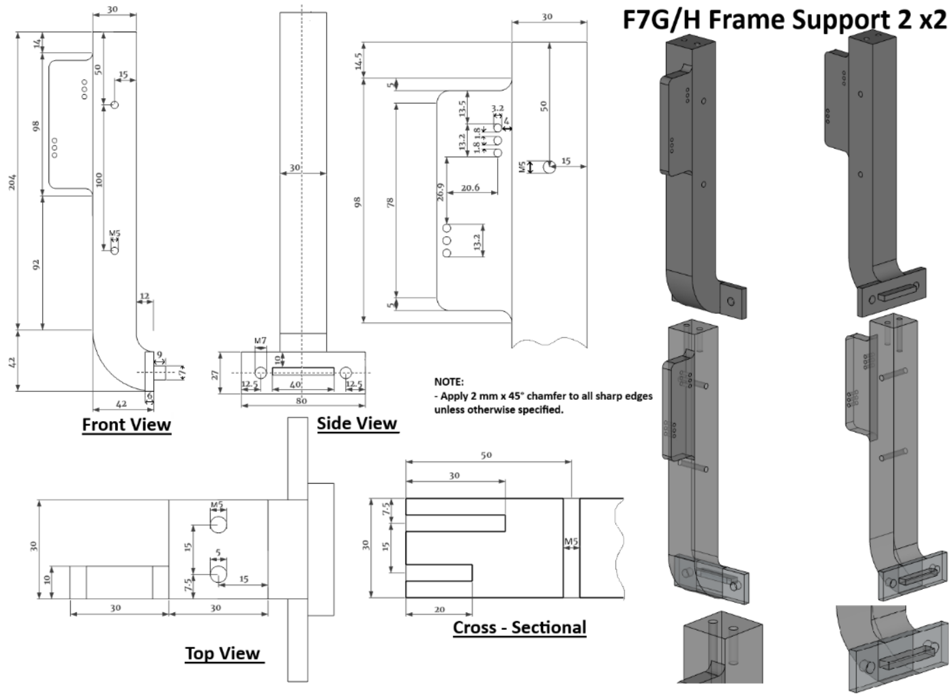
<u>Below are Older Version that was done using Microsoft Paint</u>

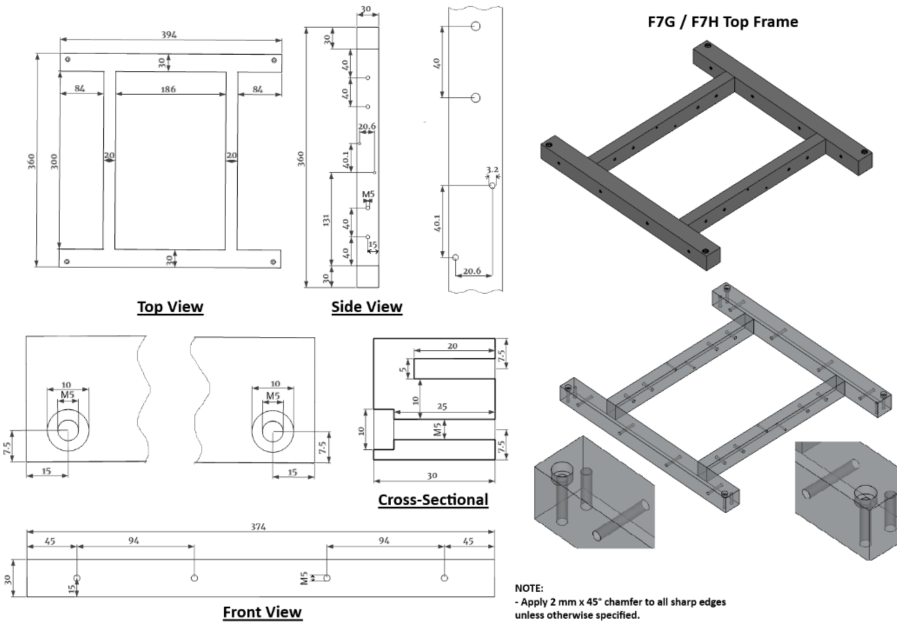




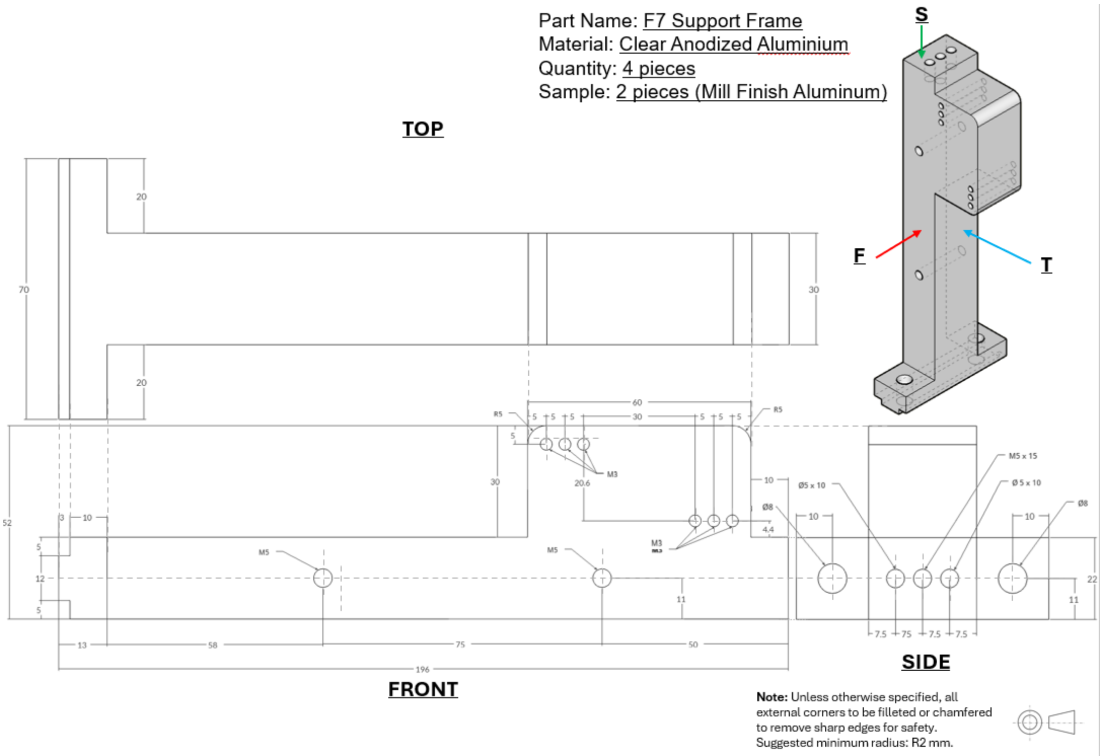

F7 Top Frame
Top View
Side View
Cross-Sectional
Front View
NOTE:
- Apply 2 mm x 45° chamfer to all sharp edges
unless otherwise specified.

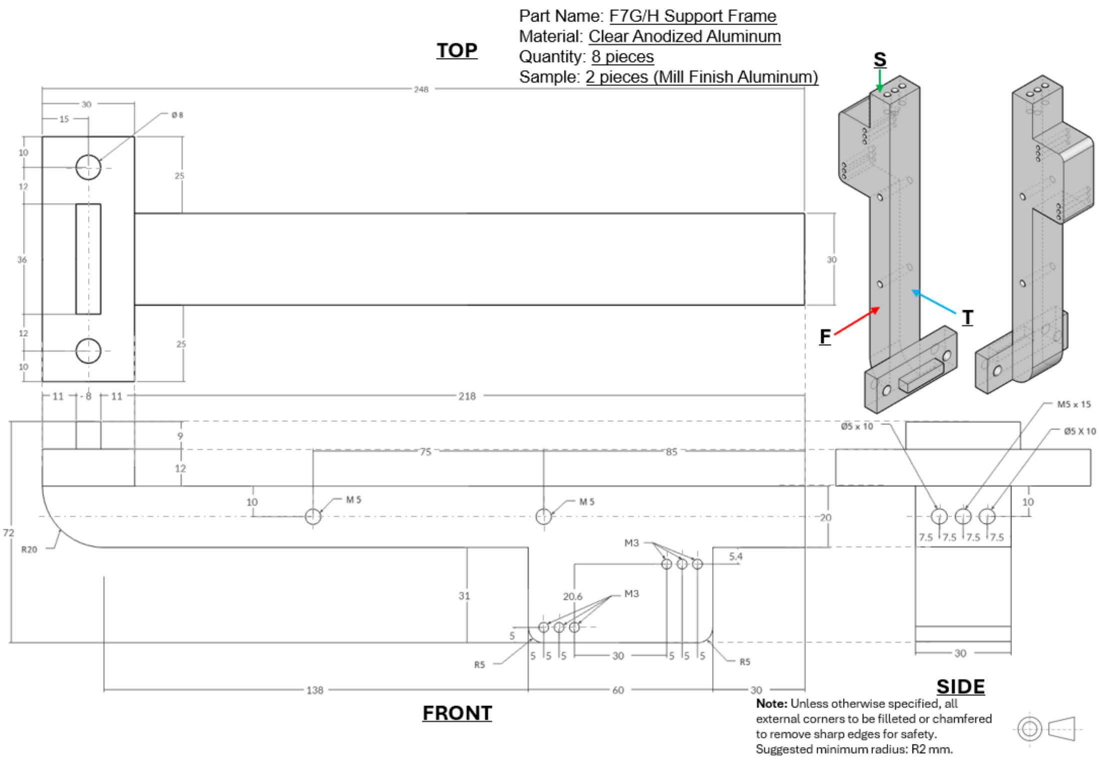

F7G/H Frame Support 1 x2
Front View
Side View
Top View
Cross - Sectional
NOTE:
- Apply 2 mm x 45° chamfer to all sharp edges
unless otherwise specified.


F7G/H Frame Support 2 x2
Front View
Side View
NOTE:
- Apply 2 mm x 45° chamfer to all sharp edges unless otherwise specified.
Top View
Cross - Sectional

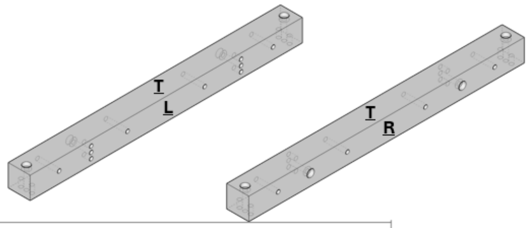

F7G / F7H Top Frame
Top View
Side View
Cross-Sectional
Front View
NOTE:
- Apply 2 mm x 45° chamfer to all sharp edges unless otherwise specified.

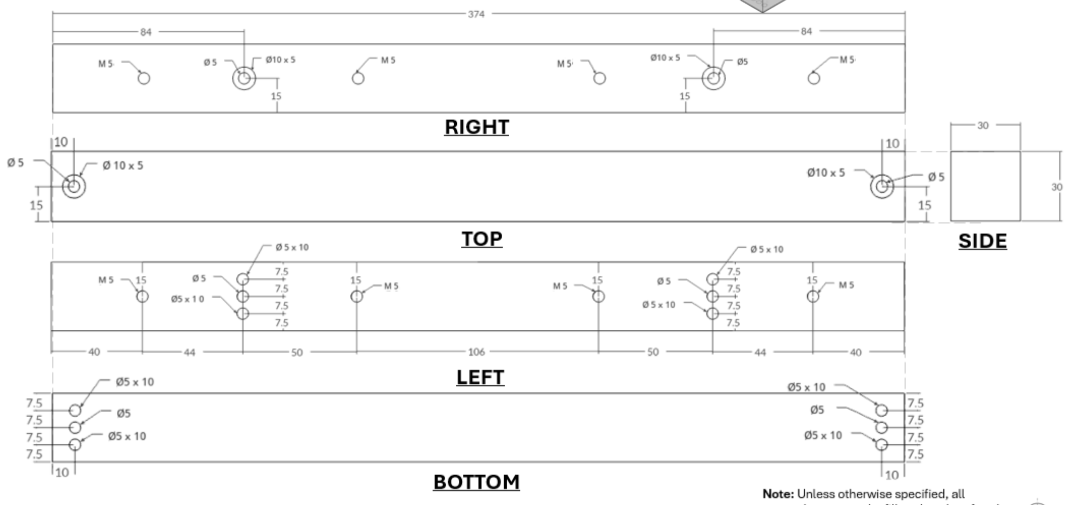
Below is the Latest Version that was done using Microsoft PowerPoint





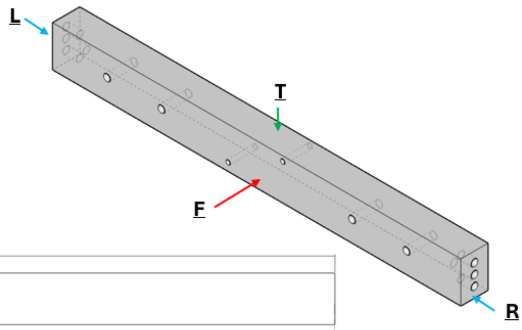
Part Name: Connector Assembly
Material: Clear Anodized Aluminum
Quantity: 4 pieces
Sample: 2 pieces (Mill Finish Aluminum)

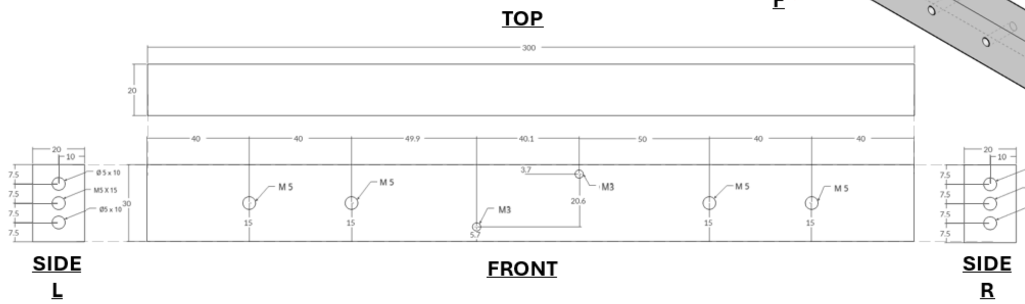




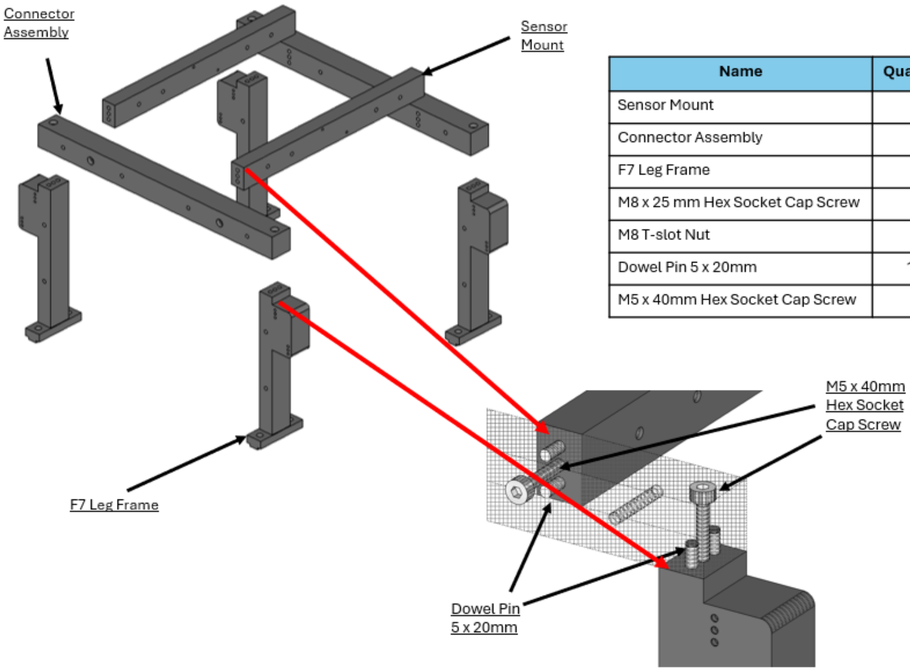
Note: Unless otherwise specified, all external corners to be filleted or chamfered to remove sharp edges for safety. Suggested minimum radius: R2 mm.

Part Name: Sensor Mount
Material: Clear Anodized Aluminum
Quantity: 4 pieces
Sample: 2 pieces (Mill Finish Aluminum)



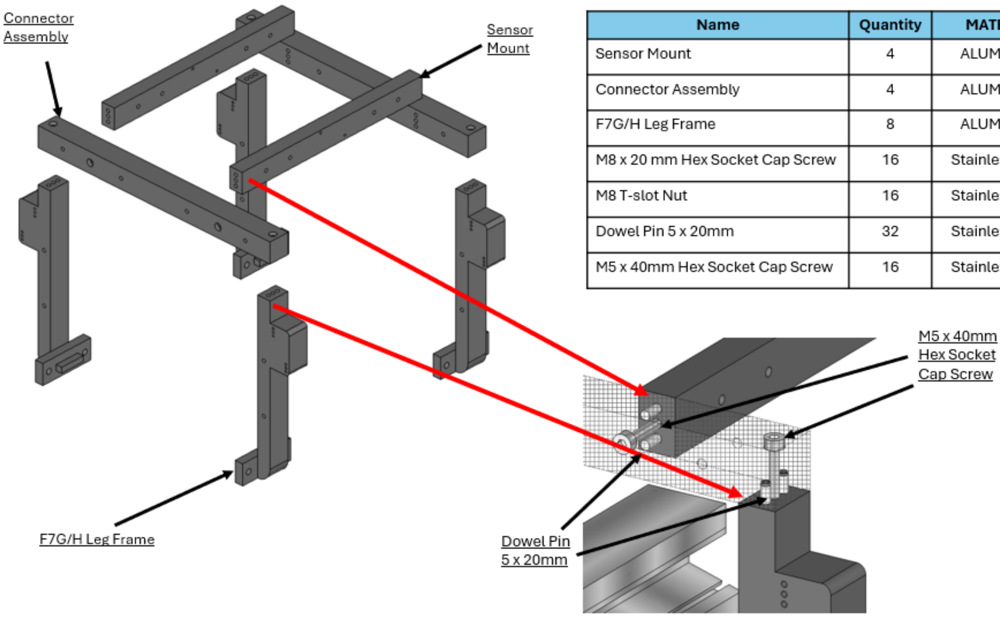


Note: Unless otherwise specified, all external corners to be filleted or chamfered to remove sharp edges for safety. Suggested minimum radius: R2 mm.



Name	Quantity	MATERIAL
Sensor Mount	2	ALUMINIUM
Connector Assembly	2	ALUMINIUM
F7 Leg Frame	4	ALUMINIUM
M8 x 25 mm Hex Socket Cap Screw	8	Stainless Steel
M8 T-slot Nut	8	Stainless Steel
Dowel Pin 5 x 20mm	16	Stainless Steel
M5 x 40mm Hex Socket Cap Screw	8	Stainless Steel

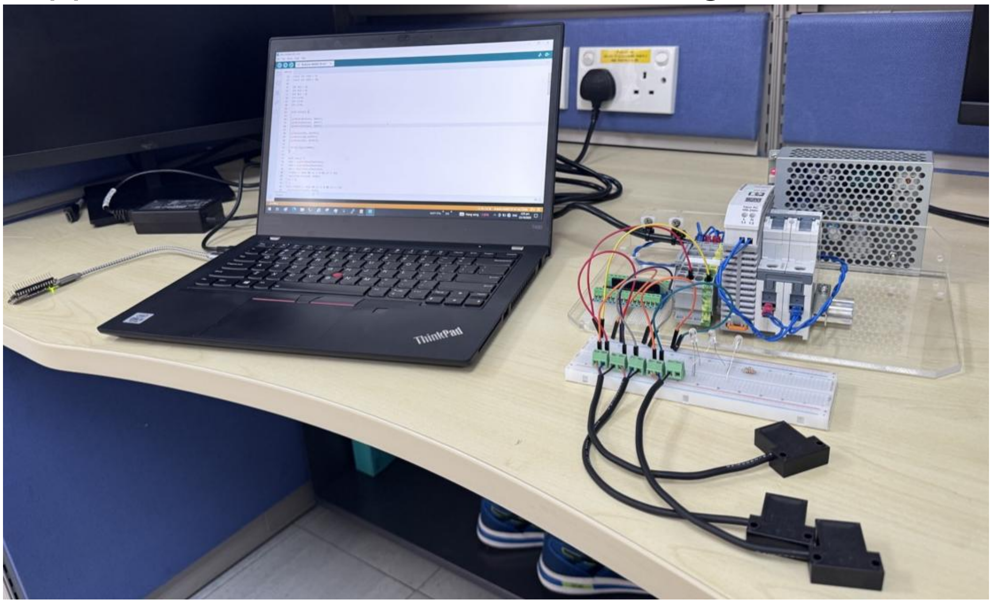
F7 Frame Assembly Overview



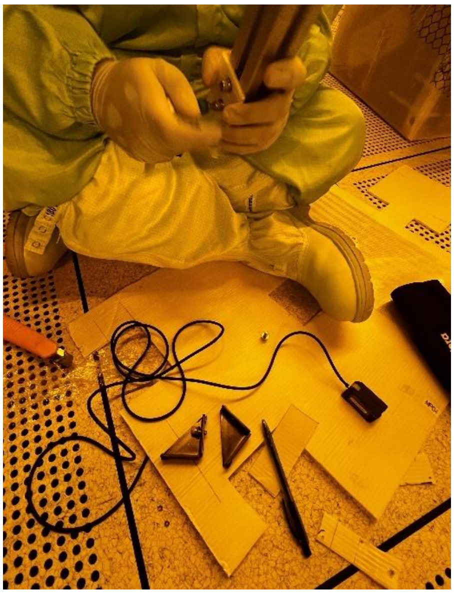
Name	Quantity	MATERIAL
Sensor Mount	4	ALUMINIUM
Connector Assembly	4	ALUMINIUM
F7G/H Leg Frame	8	ALUMINIUM
M8 x 20 mm Hex Socket Cap Screw	16	Stainless Steel
M8 T-slot Nut	16	Stainless Steel
Dowel Pin 5 x 20mm	32	Stainless Steel
M5 x 40mm Hex Socket Cap Screw	16	Stainless Steel

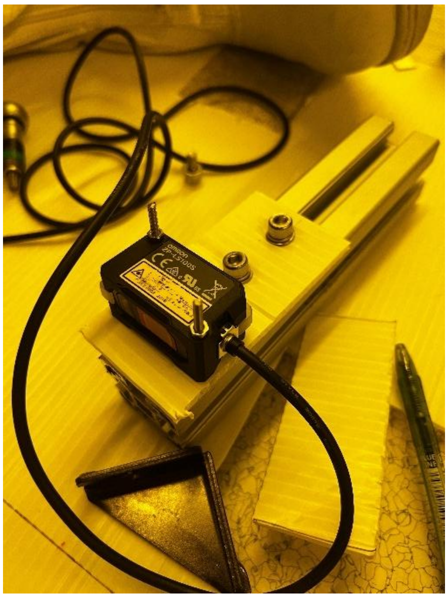
F7G/H Frame Assembly Overview

6.4 Appendix D – On-site Sensor Testing

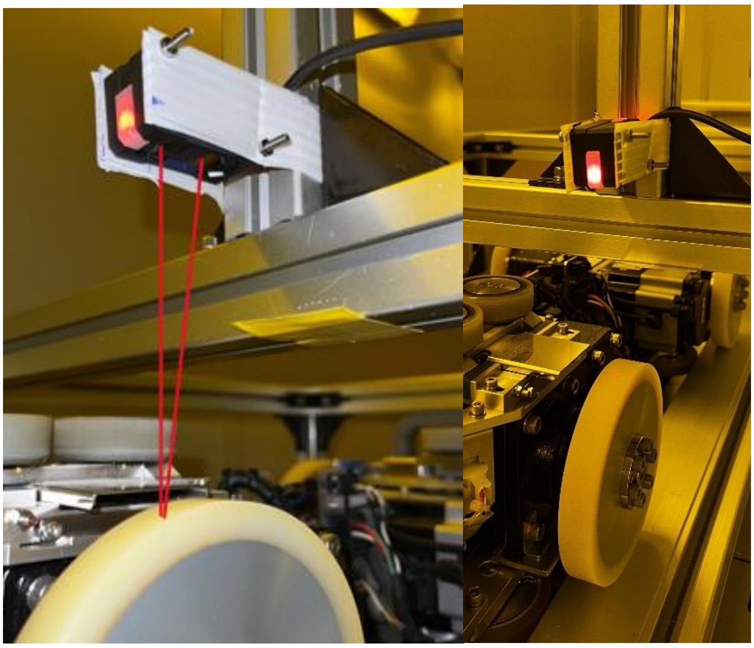


Magnetic Sensor Switch Testing

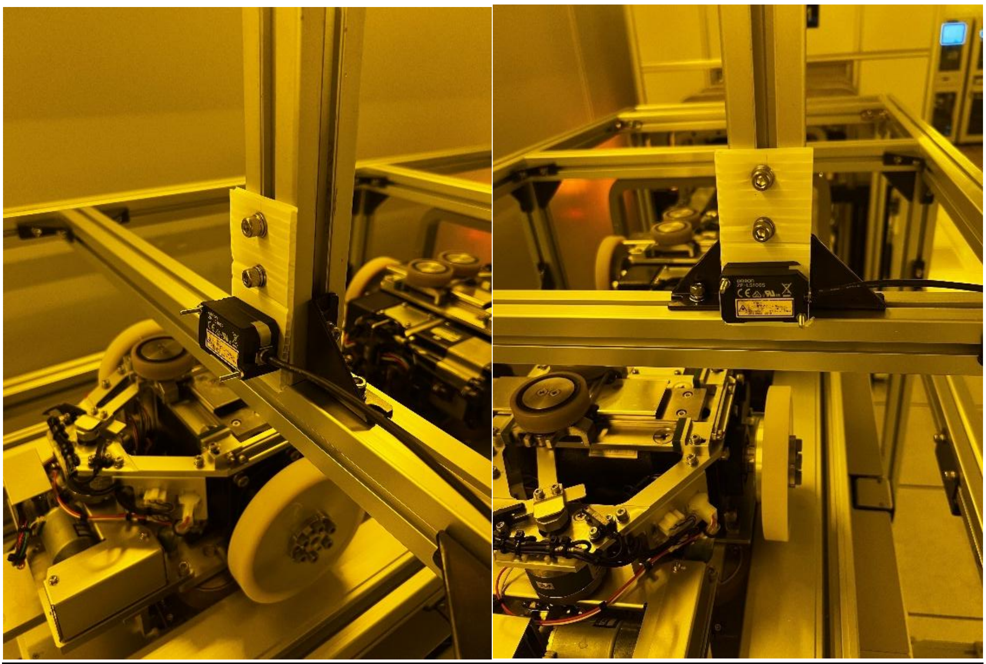




DIY of the Sensor Mount



Parallel Sensor Placement



Perpendicular Sensor Placement